

Brochosomes as an antireflective camouflage coating for leafhoppers

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The authors provide **important** insights into a system of insect camouflage where a coating of self-made nano-particles (brochosomes) reduces the reflection of UV-light leading to lower predation by spiders. **Compelling** evidence is provided by micro-UV-Vis spectroscopy, electron microscopy, transcriptome and proteome analysis, histology, in-vivo predation assays and gene knock-downs. The phylogenetic analyses provide evidence that the genes coding for the brochosome proteins are clade-specific and have diversified by gene duplication.

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Abstract

In nature, insects face immense predation pressure, where visual cues play a vital role in predators locating them. To counter this threat, insects employ a variety of nano- and microstructures on their cuticular layer to manipulate and interact with light, enhancing anti-reflective properties and providing camouflage or reducing detectability by predators. Leafhoppers have a unique extra-cuticular coating called brochosome, yet its anti-reflective functions and protein composition remain unclear. Our study demonstrates strong anti-reflective properties of brochosomes, effectively reducing reflectance on the cuticle surface, especially in the ultraviolet spectrum, to improve evasion from visual predators. Furthermore, we identify four novel structural proteins of the brochosome (BSM) for the first time. Inhibiting their synthesis by RNAi alters brochosome morphology, impacting the optical properties of the cuticle surface. Evolutionary origin analysis of BSM suggests that brochosomes likely originated from a process involving duplication–divergence. Our study reveals that leafhoppers employ a unique camouflage strategy by secreting brochosomes as anti-reflection nano-coatings, enabling them to evade natural predators and contributing to their evolutionary success.

Introduction

Predation exerts significant selective pressure on the evolution of various species, driving the development of anti-predator strategies across the animal kingdom (Galloway et al., 2020; Stevens & Ruxton, 2018). Camouflage is a widely employed strategy, allowing animals to blend into their surroundings by adjusting their coloration and patterns to reduce detection by visual predators (Hughes et al., 2019; Skelhorn & Rowe, 2016). Nevertheless, the presence of predators with multispectral vision, capable of perceiving non-visible light spectra like infrared and ultraviolet, can compromise traditional color and pattern-based camouflage (Galloway et al., 2020; Jones et al., 2007; Stevens, 2007; Thery & Gomez, 2010), paralleling the challenges of military camouflage under non-visible light (Xi et al., 2023). Consequently, achieving alignment between an animal's surface optical characteristics and its surroundings becomes a crucial requirement in visual camouflage strategies.

Insects, one of the most diverse animal groups on earth, contend with intense predation pressure (Cinel et al., 2020; Sheikh et al., 2017). Many of their predators, including birds, reptiles, and predatory arthropods, rely on visual cues, with some possessing highly developed ultraviolet (UV) vision, where UV light plays a critical role in locating insect prey (Lim & Ben-Yakir, 2020; Stobbe et al., 2008). Furthermore, the natural backgrounds where insects reside, such as leaves, tree bark, and soil, exhibit minimal light reflection (Endler, 1997). Consequently, light reflections originating from an insect's body significantly elevate its risk of exposure to predators (Tovee, 1995). To counter this challenge, the various nano- and microstructural features of the insect cuticle have excellent antireflective properties that affect insect body coloration, reduce surface light reflection and aid camouflage (Vukusic & Sambles, 2003; Watson et al., 2017). The first instances of anti-reflective structures were observed in the compound eyes of moths, the corneas of which are covered by hexagonally arranged protrusions that are approximately 200 nanometers apart (Bernhard, 1965; Blagodatski et al., 2015; Clapham & Hutley, 1973). Subsequent research has revealed that anti-reflective structures are widespread on insect body surfaces, primarily found on compound eyes and wings (Blagodatski et al., 2015; Chotard et al., 2022; Ho et al., 2016; Monteiro et al., 2019; Stavenga et al., 2005). These nano- and micro-structures on the cuticular layer are believed to possess anti-reflective properties, effectively minimizing light reflections from their bodies and thus preventing detection by predators (Blagodatski et al., 2015; Chotard et al., 2022; Ho et al., 2016; Monteiro et al., 2019; Stavenga et al., 2005).

Leafhoppers (Cicadellidae), one of the largest insect families with over 22,000 species, possess unique extra-cuticular coating known as brochosome. Brochosomes, typically hollow, honeycomb-like spheres with a diameter ranging from 0.2 to 0.6 micrometers, are synthesized in the Malpighian tubules, secreted through the hindgut, and applied as a coating on the fresh cuticle following molting (Rakitov, 2000; Rakitov, 2009). Brochosomes primarily comprise lipids and proteins, with protein content ranging from 45% to 75%, but the exact proteins responsible for brochosome formation remain unknown (Li et al., 2022; Rakitov et al., 2018; Yuan et al., 2023). Some have proposed that brochosomes may serve as a protective layer on the leafhopper's cuticle surface, with a potential role in enhancing the hydrophobicity of their cuticle surface, providing defense against pathogens and predators (Rakitov & Gorb, 2013a; Rakitov, 2009). Apart from these known function of hydrophobicity and the shedding of brochosomes in helping leafhopper escape spider webs, there is currently no definitive biological evidence supporting other hypothesized functions of brochosomes (Lin et al., 2021; Rakitov & Gorb, 2013a, 2013b). Furthermore, it has been suggested by certain researchers that the brochosomes present on the surface of leafhoppers may play a role in their camouflage abilities (Swain, 1936). In recent years, studies investigating bio-inspired materials that mimic the structure of brochosomes have demonstrated remarkable antireflective properties (Lei et al.,

2020 [\[1\]](#); Yang et al., 2017 [\[2\]](#)), and further elucidated the antireflective mechanisms based on the brochosome geometry (Wang et al., 2024 [\[3\]](#)). These findings offer valuable insights, suggesting that leafhoppers may employ a distinctive strategy, distinct from other insects, to achieve anti-reflection properties on their surfaces. Consequently, we propose the hypothesis that brochosomes may function as a camouflage coating on the leafhopper's cuticle surface, aiding in the reduction of surface optical reflections and enabling the leafhopper to remain concealed, thereby evading detection by visual predators and ultimately escaping predation.

In this study, we combined spectrophotometry, electron microscopy, transcriptome analysis, proteome analysis, gene function validation, and leafhopper and jumping spider bioassays to characterize the protein composition and the function of brochosome in leafhopper camouflage. We focused our research on the rice green leafhopper, *Nephotettix cincticeps*, a well-known agricultural pest with an extensive research history (Hibino, 1996 [\[4\]](#); Wei et al., 2018 [\[5\]](#); Yan et al., 2021 [\[6\]](#)). Our results revealed the pronounced effectiveness of brochosome coverage in reducing light reflection on leafhoppers *N. cincticeps*, especially within the scope of the ultraviolet spectrum. By conducting RNAi experiments, we successfully identified four brochosome structural protein for the first time. The suppression of their synthesis induced changes in brochosome morphology, influencing the optical characteristics of the leafhopper's cuticle surface. Furthermore, our analysis of the evolutionary origin of brochosomes indicates that they likely originated through a process involving duplication–divergence. Our findings support brochosomes as a unique camouflage coating that enables leafhoppers to evade visual predators.

Results

Brochosomes are a distinctive coating on the cuticle surface of leafhopper *N. cincticeps*

The cuticle surface of rice green leafhopper *N. cincticeps* is coated with brochosomes (Figure 1A [\[7\]](#) and B [\[8\]](#)), which are approximately 350nm in diameter and featuring a hollow sphere with a honeycomb-like structure (Figure 1C [\[9\]](#) and D [\[10\]](#)). The leafhopper *N. cincticeps* possesses two pairs of Malpighian tubules, each consisting of a proximal, distal, and terminal segment (Figure 1E [\[11\]](#)). Brochosomes are synthesized in the distal segment of the Malpighian tubule, which exhibits a swollen and rod-shaped appearance (Figure 1E [\[11\]](#)). The epithelial cells in this segment contain large spherical nuclei, an extensive rough endoplasmic reticulum, and multiple Golgi regions (Figure 1E [\[11\]](#) and F [\[12\]](#)). Within the Golgi-derived vacuole, the brochosomes undergo progressive development, transforming their initially spherical granules into a honeycomb-like surface with closely spaced invaginations (Figure 1G–I [\[13\]](#)). Mature brochosomes are stored in secretory vacuoles at the boundaries of the cells and are subsequently secreted into the lumen of the Malpighian tubules (Figure 1F [\[12\]](#)).

The optical properties of the leafhopper's cuticle surface are intricately connected to the brochosome coating

To investigate the interplay between brochosome coverage and the optical features of *N. cincticeps* cuticle surface, we observed the relationship for male and female adults at 5, 10, 15, 20, and 25 days post-eclosion. *N. cincticeps* exhibited a gradual shift in body color with increasing eclosion time, manifesting clear distinctions between males and females (Figure 2A-figure supplement 1A and B). Males transformed from light green to dark green, while females progressively transitioned from light green to translucent, displaying an iridescence (Figure 2A-figure supplement 1A and B). Furthermore, the brightness of the leafhopper cuticle surface under UV

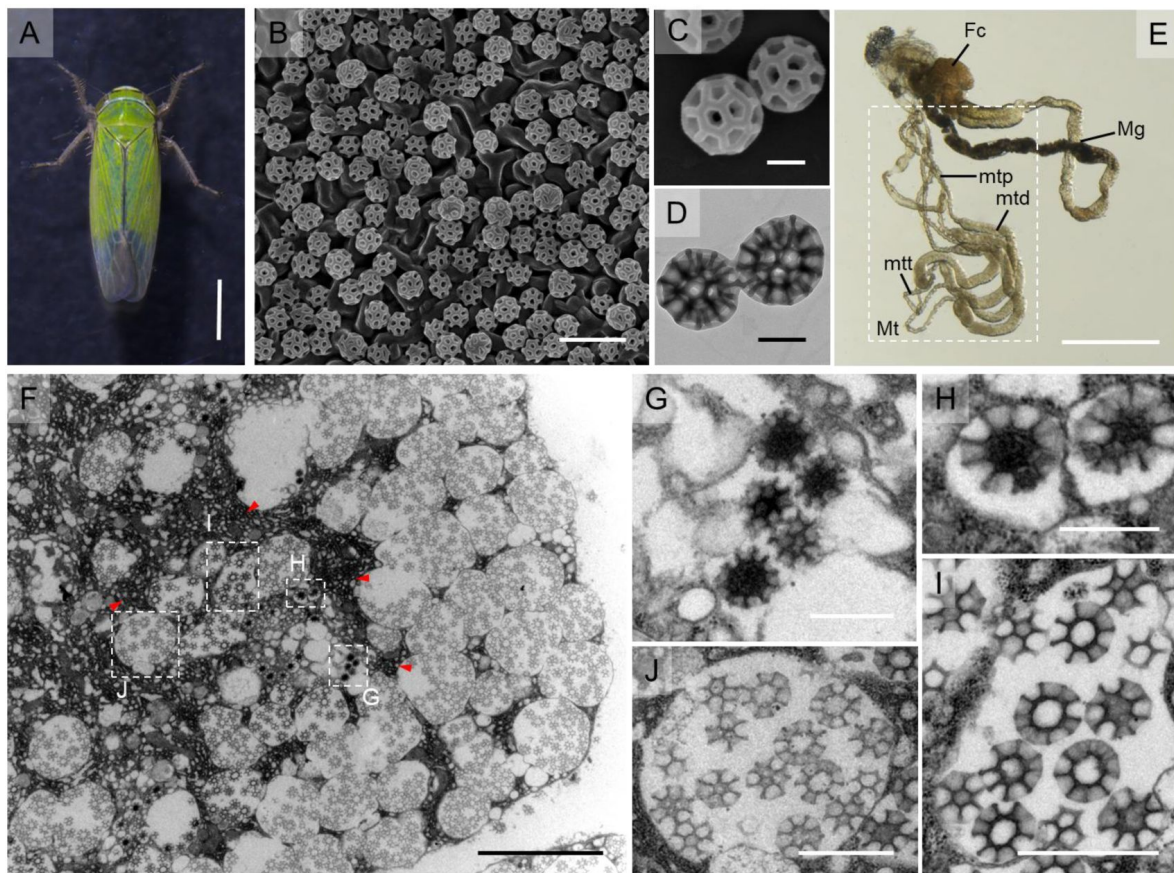


Figure 1.

Brochosomes are a distinctive coating on the cuticle surface of leafhopper.

(A) Male adult of the green rice leafhopper *N. cincticeps*. Bar, 1 mm. (B) Brochosomes on the surface of the forewing of *N. cincticeps*. Bar, 1 μ m. (C, D) The morphologies of brochosomes by SEM (C) and TEM (D). Bar, 200 nm. (E) Alimentary tract and the Malpighian tubules of *N. cincticeps*. The leafhopper *N. cincticeps* have two pairs of Malpighian tubules, each divided into proximal segment, distal segment, and terminal segment. The brochosomes are synthesized in the distal segment, which is dilated and rod-shaped. Bar, 1 mm. (F) The distal segment epithelial cell displays an extensive rough endoplasmic reticulum and multiple Golgi regions (red arrow) containing developing brochosomes in its basal portion, as well as a number of secretory vacuoles with mature brochosomes near the cell border. Bar, 5 μ m. (G) The initial stage of the development of brochosome. Bar, 500 nm. (H) The two brochosomes that are developing inside the primary vesicles are shown in close-up. Regular invaginations appear on the surface of the growing brochosome at the same time that its matrix separates into a looser core and a denser wall. Bar, 500 nm. (I) Larger vesicles containing numerous BS, formed by the fusion of multiple primary vesicles. The surface of the brochosomes has regular cell-like invaginations, and it is surrounded by amorphous flocculent material with a moderate electron density. Bar, 1 μ m. (J) A vesicle filled with mature brochosomes, each mature brochosome has a spherical inner cavity and a well-swollen outer margin of the septa. Bar, 1 μ m. E, F, G, H are the enlargement of boxed area in D. Fc, filter chamber; Mg, midgut; Mt, Malpighian tubules; mtd, distal segment of the Malpighian tubule; mtp, proximal segment of the Malpighian tubule; mtt, terminal segment of the Malpighian tubule. All images are representative of at least three replicates.

light intensified with the extension of post-eclosion time, and the UV reflectance of female cuticle surface was significantly higher than that of males after 15 days post-eclosion (Figure 2B-figure supplement 1A and B).

To validate the correlation between brochosome coverage, body color, and UV light reflectance, we examined the distribution of brochosomes on the cuticle surface of male and female *N. cincticeps* at various post-eclosion intervals using scanning electron microscopy (SEM). SEM results unveiled a gradual reduction in brochosome coverage on the cuticle surface of both sexes as post-eclosion time progressed. Specifically, male coverage decreased from 90% to approximately 40%, whereas female coverage descended from 90% to around 10% (Figure 2C-figure supplement 1A-C). Microscopic analyses of the Malpighian tubule revealed that, with prolonged post-eclosion time, the distal segment of the tubule underwent gradual atrophy, particularly pronounced at 15 days post-eclosion. Notably, females exhibited a more marked atrophy in the distal segment compared to males. By 25 days post-eclosion, the distal segment of the Malpighian tubule in females became indistinguishable from other segments (Figure 2D-figure supplement 1A and B). This denotes a progressive reduction in post-eclosion brochosome synthesis, resulting in a corresponding decrease in brochosome coverage on the cuticle surface. Furthermore, the distribution of brochosomes on the leafhopper cuticle surface may be correlated with its optical characteristics. To substantiate this hypothesis, we analyzed the light reflection characteristics of leafhopper forewings at various post-eclosion time points. The findings revealed a positive correlation between brochosome coverage on the forewings and light reflection values, underscoring the remarkable anti-reflective attributes of the brochosome coating on leafhopper cuticle surface (Figure 2E-figure supplement 1D and E). Brochosomal coverage was found to significantly decrease the reflectance of ultraviolet light on the leafhopper's surface, reducing it from approximately 30% to 20%. Additionally, the reflectance of visible and infrared light was also notably diminished, dropping from around 20% to 10% (Figure 2E-figure supplement 1D and E). In summary, the presence of brochosomal coverage resulted in an overall reduction of surface light reflection by approximately 30%, highlighting its substantial antireflective properties.

Preferential capture of leafhoppers with fewer brochosomes by jumping spiders

Given that UV light serves as a crucial visual cue for arthropod predators, especially common visual hunters like jumping spiders (Cronin & Bok, 2016 [\[4\]](#); Li & Lim, 2005 [\[5\]](#); Morehouse et al., 2017 [\[6\]](#); Silberglied, 1979 [\[7\]](#); Zou et al., 2011 [\[8\]](#)), the function of the brochosome coating on leafhoppers' cuticle surface could be to avoid predation by reduction of UV light reflectance. This prompted us to investigate the effect of brochosome coatings on reducing predation risk in leafhoppers. Predation experiments were conducted using *Plexippus paykulli* (Figure 2F-figure supplement 2), a common jumping spider in rice fields known to prey on leafhoppers (Yang et al., 2018 [\[9\]](#)). We observed the spiders' feeding behavior, noting the time of the first attack and the targeted leafhopper, using these metrics to evaluate predation efficiency (Taylor et al., 2014 [\[10\]](#); Walker & Rypstra, 2002 [\[11\]](#)). In the experimental group comprising females and males at 5 days post-eclosion, *P. paykulli* did not exhibit a clear predation preference (Figure 2G [\[12\]](#)). However, in the experimental group involving females and males at 25 days post-eclosion, there was a higher probability of predation on female *N. cincticeps* compared to male ones (Figure 2H [\[13\]](#)). For male *N. cincticeps* in both the 5-day and 25-day post-eclosion experimental groups, *P. paykulli* showed a stronger inclination towards preying on individuals that were eclosed for 25 days, resulting in significantly shorter predation times compared to those at 5 days post-eclosion (Figure 2I [\[14\]](#)). These patterns were consistently observed for female leafhoppers in both the 5-days and 25-days post-eclosion (Figure 2J [\[15\]](#)). The cumulative findings strongly suggest a distinct preference of the jumping spider *P. paykulli* for individuals with lower brochosome coverage on the cuticle surface. Based on these experiments, we proposed the hypothesis that the brochosome coating on the leafhopper cuticle surface may mitigate predation risk by reducing surface light reflection, particularly in the UV spectrum.

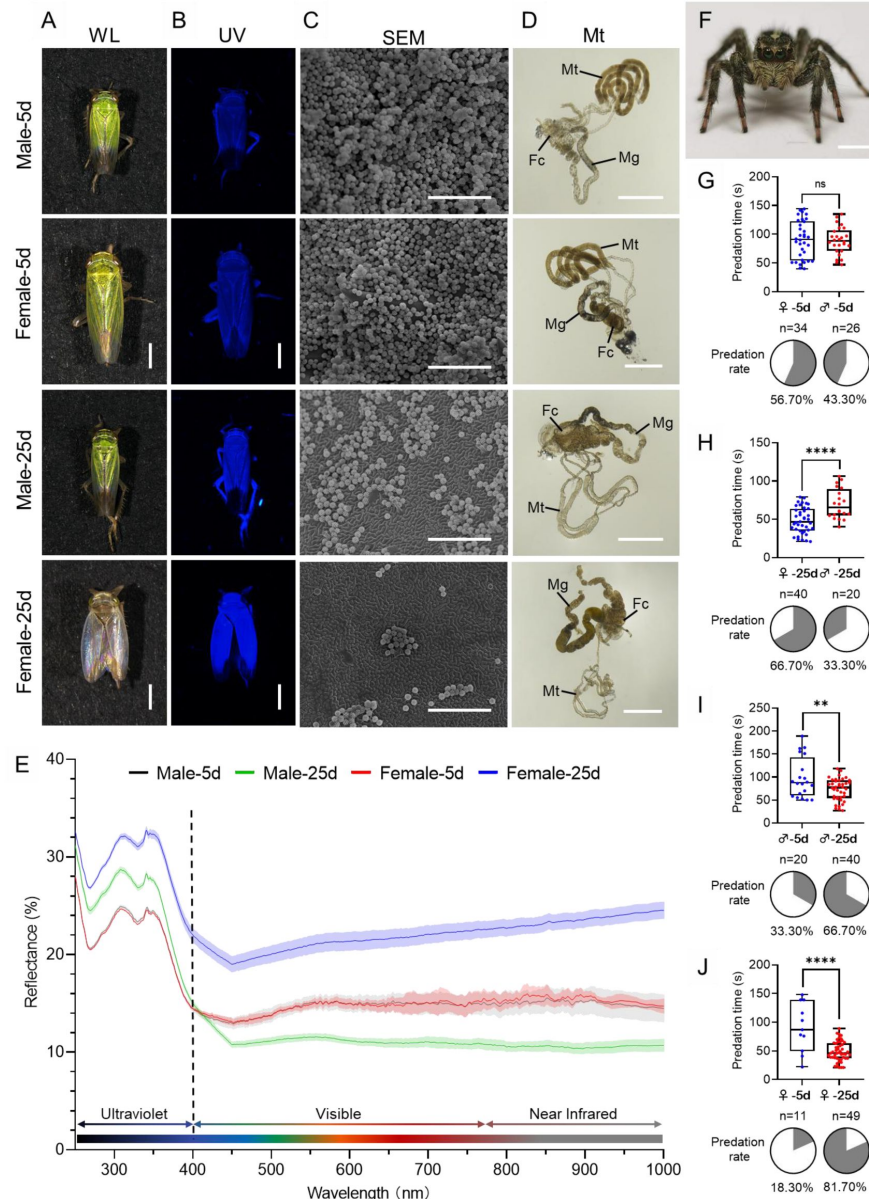


Figure 2.

The distribution of brochosomes on the cuticle surface of leafhopper *N. cincticeps* are associated with predation by jumping spiders.

(A, B) Images of leafhopper *N. cincticeps* males and females in white (A) and ultraviolet light (B) at 5 and 25 days post-eclosion, respectively. Bar, 1 mm. (C, D) The distribution of brochosomes on the surface of a forewing (C) and morphological changes of the Malpighian tubules (D) of *N. cincticeps* males and females at 5 and 25 days post-eclosion. Bar, 5 μ m in C; Bar, 1 mm in D. (E) Reflectance spectra of female and male forewing of *N. cincticeps* at 5 and 25 days post-eclosion. (F) Images of the jumping spider *P. paykulli*. Bar, 2 mm. (G-J) Jumping spiders prefer leafhoppers with little brochosome covering as food. In predation experiment, jumping spiders offered *N. cincticeps* male and female at 5 days post-eclosion (G), male and female at 25 days post-eclosion (H), males at 5 and 25 days post-eclosion (I), and females at 5 and 25 days post-eclosion (J). Data on predation times are displayed using the traditional box and whisker shapes. All box plots with whiskers represent the data distribution based on five number summary statistics (maximum, third quartile, median, first quartile, minimum), each dot in box plot represents an independent experiment. ** $P < 0.01$, **** $P < 0.0001$, ns no significance, Statistical significance was determined by unpaired t test with Welch's correction method. Predation preference is shown in the pie chart. All images are representative of at least three replicates.

Identifying the major structural proteins of brochosomes

Despite the initial discovery of brochosomes in the early 1950s and the confirmation in the 1960s that they are protein-lipid particles (Gouranton & Maillet, 1967 [\[1\]](#)), the specific protein composition of brochosomes remains unknown. Recent studies suggested that they are primarily composed of brochosomins (BSM) and brochosome-associated proteins (BSAP) (Li et al., 2022 [\[2\]](#); Rakitov et al., 2018 [\[3\]](#)). BSM, a novel class of secreted proteins with molecular weights ranging from 21 to 40 kDa, is considered as the major structural component of brochosomes (Rakitov et al., 2018 [\[3\]](#)). Proteomics studies have identified the proteins composing Brochosomes, but the specific proteins involved in their formation remain unclear.

To further elucidate the essential protein components of brochosomes, transcriptomic and proteomic data derived from brochosome of *N. cincticeps* were comprehensively analyzed. Based on our prior integrated analysis of the brochosome transcriptome and proteome (Wu et al., 2023 [\[4\]](#)), in conjunction with literature search results, we selected 50 candidate genes for functional analysis using RNAi-mediated gene silencing. Through RNAi experiments, we successfully identified four genes encoding brochosome structural proteins that led to morphological changes in brochosomes (**Figure 3A** [\[5\]](#)). According to the coding sequence (CDS) length of these genes, the four proteins were named as BSM-1 to BSM-4 (GenBank accession number PP273097, PP273098, PP273099, PP273100) (**Figure 3B** [\[6\]](#)). Homology analysis suggested that BSM-coding genes might be paralogous (Figure supplement 3A). Although BSM2 and BSM3 exhibited low sequence homology, their protein structures were highly conserved (Figure supplement 3B-C). To validate the functions of these four BSM-coding genes, we synthesized dsRNA from two non-overlapping regions of each gene to minimize off-target effects (Figure supplement 4A). RT-qPCR revealed that both individual and mixed dsRNA injections effectively suppressed the expression of the BSM-coding genes, with mixed injections achieving higher efficiency (Figure 3C-F-figure supplement 4C-J). SEM observations showed that both injection methods induced significant morphological changes in brochosomes, characterized by increased diameters and deformed honeycomb-like structures (Figure 3A-figure supplement 4B). Statistical analysis of SEM data from mixed dsRNA injections indicated a 60-70% reduction in brochosome distribution area and a 20% incidence of morphologically abnormal brochosomes compared to dsGFP controls (Figure supplement 5A-B). Additionally, temporal and spatial expression analyses demonstrated that the BSM-coding genes were specifically expressed in the Malpighian tubules (**Figure 3G-J** [\[7\]](#)) and exhibited relatively stable expression during the early post-eclosion period, followed by a gradual decline after 10 days (**Figure 3K-N** [\[8\]](#)). By 25 days post-eclosion, the expression of BSM-coding genes declined to around 10% in females and 30-40% in males, respectively (**Figure 3K-N** [\[8\]](#)). This phenomenon is consistent with our initial microscopic observations, suggesting that the gradual reduction in brochosome synthesis contributes to the decrease of brochosome coverage on the cuticle surface of *N. cincticeps* after adulthood.

Brochosome coating diminishes light reflection and facilitates predator avoidance

To further investigate the function of brochosomes in leafhopper cuticle surface associated with light reflection and predator avoidance, we implemented RNAi to simultaneously suppress the expression of all four BSM-coding genes with dsBSM mixture injection. RT-qPCR results demonstrated a substantial downregulation in the expression of these genes compared to the dsGFP control (**Figure 4F** [\[9\]](#)). Notably, under ultraviolet light, both male and female leafhoppers exhibited a significant increase in UV reflection on their cuticle surface following dsBSM treatment (**Figure 4A** [\[10\]](#)). Spectral data corroborated these findings, showing elevated emission values in the ultraviolet spectrum on the leafhopper's cuticle surface (**Figure 4G** [\[11\]](#)). Scanning electron microscopy showed that dsBSM treatment significantly altered brochosome morphology

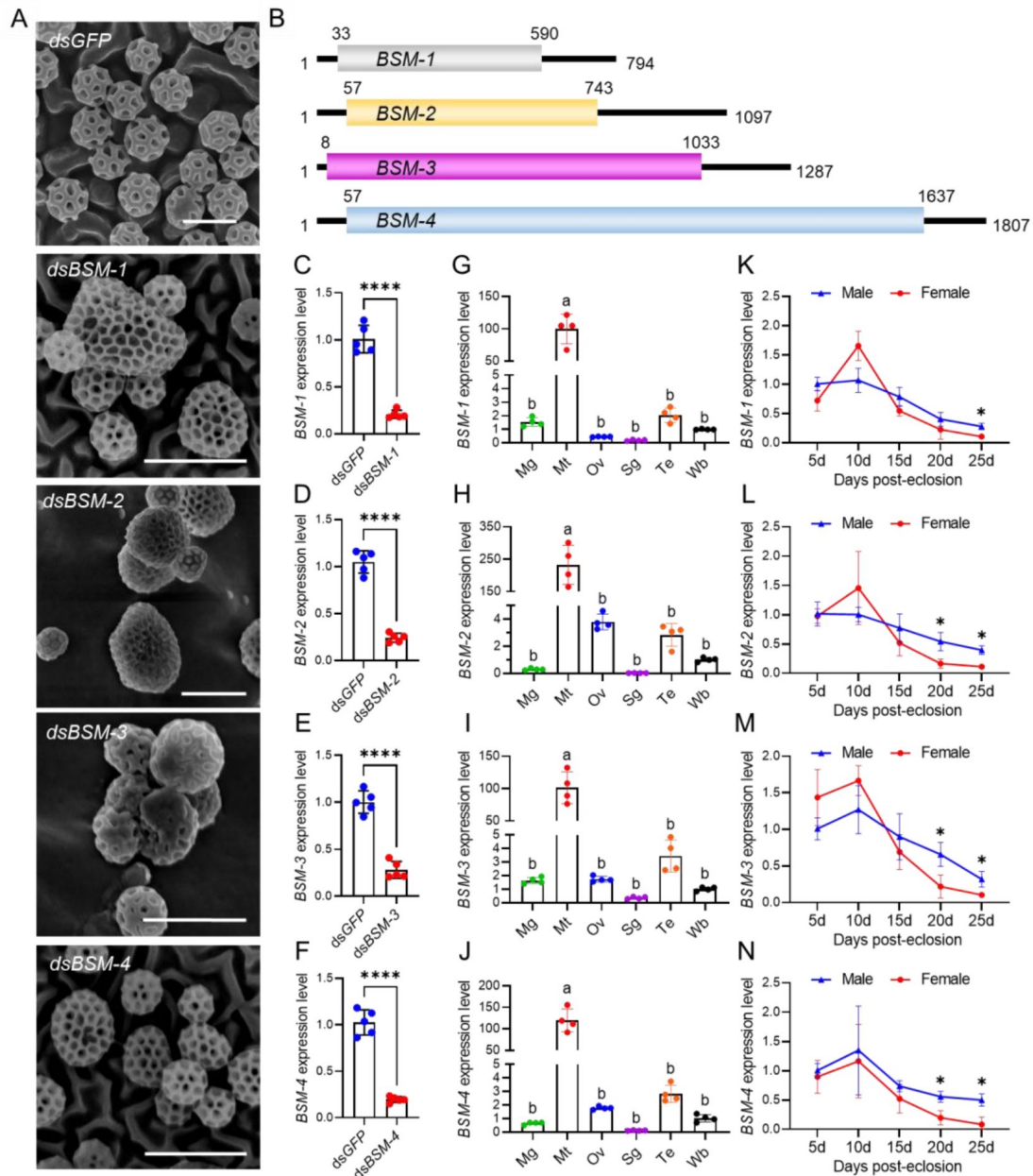


Figure 3.

Identification of brochosome structural proteins.

(A) Morphology of brochosomes on the forewing of leafhopper *N. cincticeps* at 7 days post-microinjection with dsRNA mix targeting two non-overlapping regions of each BSM gene. Bar, 500 nm. (B) Gene structures of *BSM-1*, *BSM-2*, *BSM-3* and *BSM-4*. (C-F) Transcription levels of *BSM-1* (C), *BSM-2* (D), *BSM-3* (E), and *BSM-4* (F) at 7 days post-microinjection with dsRNA mix targeting two non-overlapping regions of each BSM gene. Each data point represents the result of one independent experiment. (G-J) The abundance of BSM transcripts in different tissues and whole bodies of *N. cincticeps* was determined by RT-qPCR. Notably, *BSM-1* (G), *BSM-2* (H), *BSM-3* (I), and *BSM-4* (J) exhibited specific expression in the Malpighian tubules. Each data point represents the result of one independent experiment. (K-N) The expression patterns of *BSM-1* (K), *BSM-2* (L), *BSM-3* (M) and *BSM-4* (N) transcripts were examined in male and female leafhopper at 5, 10, 15, 20, and 25 days post-eclosion. Results were obtained from 3 independent experiments. For C-N, data shown are mean $\pm$ SD values. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$; ns no significance (C-F, two tailed Student's t-test; G-J, one-way ANOVA; K-N, two-way ANOVA). All images are representative of at least three replicates.

and distribution (**Figure 4B** [↗](#) and C-figure supplement 6A and B). The distribution area decreased by 80%, and deformed brochosomes accumulated, contrasting with the uniform distribution in the dsGFP treatment (Figure supplement 5A). Additionally, nearly 30% of brochosomes exhibited significant morphological changes following dsBSM treatment (Figure supplement 5B). Transmission electron microscopy of the Malpighian tubules indicated numerous Golgi-derived vacuoles without brochosome distribution in epithelial cells after dsBSM treatment (**Figure 4D** [↗](#) and E-figure supplement 6C and D). Predation experiments with jumping spiders revealed an increased predation preference for leafhoppers treated with dsBSM. For male leafhoppers, the proportion attacked of the predator was 70%, with a significantly shorter predation time (71.3 s) compared to the dsGFP control (100.4 s) (**Figure 4H** [↗](#)). Similarly, for females, there was a higher predation rate (68.3%) and a significantly shorter predation time (79.6 s) for dsBSM treated groups compared to the dsGFP ones (30% and 109.9 s, respectively) (**Figure 4I** [↗](#)).

Normal structure of the brochosome correlates with its antireflective properties

To further elucidate the correlation between brochosome morphology and its optical performance, we collected brochosomes from the wings of leafhoppers treated with dsGFP and dsBSM, respectively. Subsequently, these brochosomes were applied to the wings of brown planthoppers and quartz slides. SEM observations revealed that the coverage of brochosomes derived from both dsGFP and dsBSM treatments on the wings of brown planthoppers and quartz slides were nearly identical (Figure supplement 7). Spectral measurements indicated that the application of brochosomes from both dsGFP and dsBSM treatments effectively reduced the reflectance of the brown planthopper wings and quartz slides. It should be noted that brochosomes from the dsGFP treatment exhibited significantly higher anti-reflective performance compared to those from the dsBSM treatment (**Figure 4J** [↗](#) and **K** [↗](#)). Additionally, purified BSM proteins applied to quartz glass did not show improved anti-reflective performance over purified GST protein (Figure supplement 8), indicating a strong correlation between brochosome geometry and optical performance.

Brochosomes as a camouflage coating for the leafhoppers in the family Cicadellidae

Based on the forementioned findings, brochosomes of *N. cincticeps* can be considered as an anti-reflective stealth coating against visual recognition by the predator *P. paykulli*. However, it remains unknown whether brochosomes also serve as a stealth coating in other leafhopper species. Therefore, we extended our investigation to include additional leafhopper species in the family Cicadellidae, including *Recilia dorsalis*, *Empoasca onukii*, and *Psammotettix alienus*. We carefully removed the brochosomes from the forewings of these leafhopper species using acetone, and found a significant enhancement in UV reflectance on the forewings of the leafhoppers (**Figure 5A** [↗](#) and **B** [↗](#)). In addition to *N. cincticeps*, we observed a considerably higher brightness of acetone-treated forewings compared to the untreated ones under UV light for *R. dorsalis*, *E. onukii*, and *P. alienus* (**Figure 5A** [↗](#)). This result suggests that brochosomes as an anti-reflective stealth coating may be commonly existed in various species of the family Cicadellidae.

Previous research indicated that the brochosome is a unique secretion in Cicadellidae (Rakitov & Gorb, 2013b [↗](#); Rakitov et al., 2018 [↗](#)). To investigate the conservation of genes linked to brochosome synthesis in Cicadellidae, we systematically screened these four BSM coding genes across 116 other hemipteran insect species (Supplementary Table 2). Results of homology analysis reveal that BSM coding genes are orphan genes restricted to the clade of Membracoidea. However, there are variations in the species distribution of different BSM coding genes. In Membracoidea species, homologous genes for BSM3 and 4 are observed, while BSM2 is identified in Cicadellidae insects, and BSM1 in Deltocephalinae insects (**Figure 5C** [↗](#)). An intriguing observation is that in

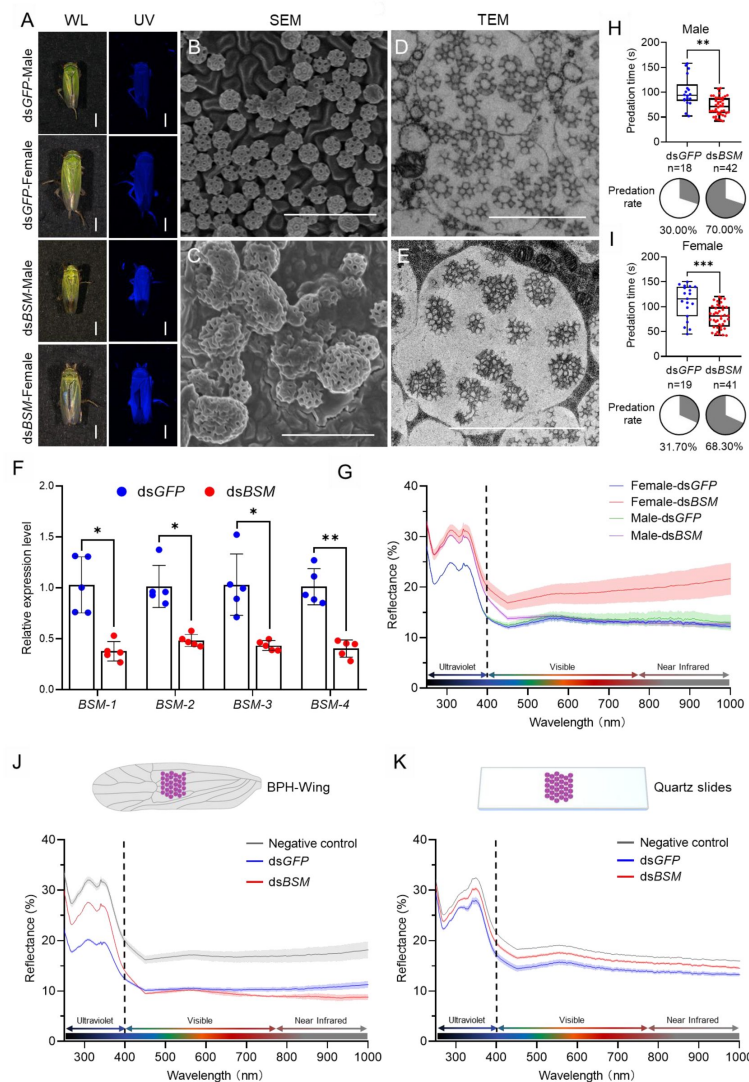


Figure 4.

RNAi inhibits brochosome synthesis, alters brochosome morphology, and influences predation in jumping spiders.

(A) Images of leafhopper *N. cincticeps* males and females in white and ultraviolet light after *dsGFP* or *dsBSM* treatment, respectively. Bar, 1 mm. (B, C) Morphology of the brochosome on the forewings of leafhoppers after *dsGFP* (B) and *dsBSM* (C) treatment. Bar, 2 μ m. (D, E) Morphology of the brochosome in the distal segment epithelial cells of Malpighian tubules after *dsGFP* (D) and *dsBSM* (E) treatment. Bar, 2 μ m. (F) The transcript levels of *BSM-1*, *BSM-2*, *BSM-3* and *BSM-4* at 7 days after *dsGFP* and *dsBSM* treatment. Each data point represents the outcome of an individual independent experiment. The presented data are expressed as mean $\pm$ SD values. Statistical significance is denoted as * $P < 0.05$ and ** $P < 0.01$, determined by two-way ANOVA. (G) Reflectance spectra of female and male forewing of *N. cincticeps* at 7 days after *dsGFP* and *dsBSM* treatment. (H and I) Jumping spiders prefer to prey on *dsBSM*-treated leafhoppers. Predation efficiency and preference of jumping spiders on males (H) and females (I) *N. cincticeps* after *dsGFP* and *dsBSM* treatment in the predation experiment. Data on predation times are displayed using the traditional box and whisker shapes. All box plots with whiskers represent the data distribution based on five number summary statistics (maximum, third quartile, median, first quartile, minimum), each dot in box plot represents an independent experiment. ** $P < 0.01$, *** $P < 0.001$, Statistical significance was determined by unpaired t test with Welch's correction method. Predation preference is shown in the pie chart. (J, K) The morphology of brochosomes is related to their optical performance. Collect brochosomes treated with *dsGFP* and *dsBSM* separately, apply them to brown planthopper wings (J) and quartz slides (K), and set up quartz slides or brown planthopper wings treated solely with acetone as negative controls. All images are representative of at least three replicates.

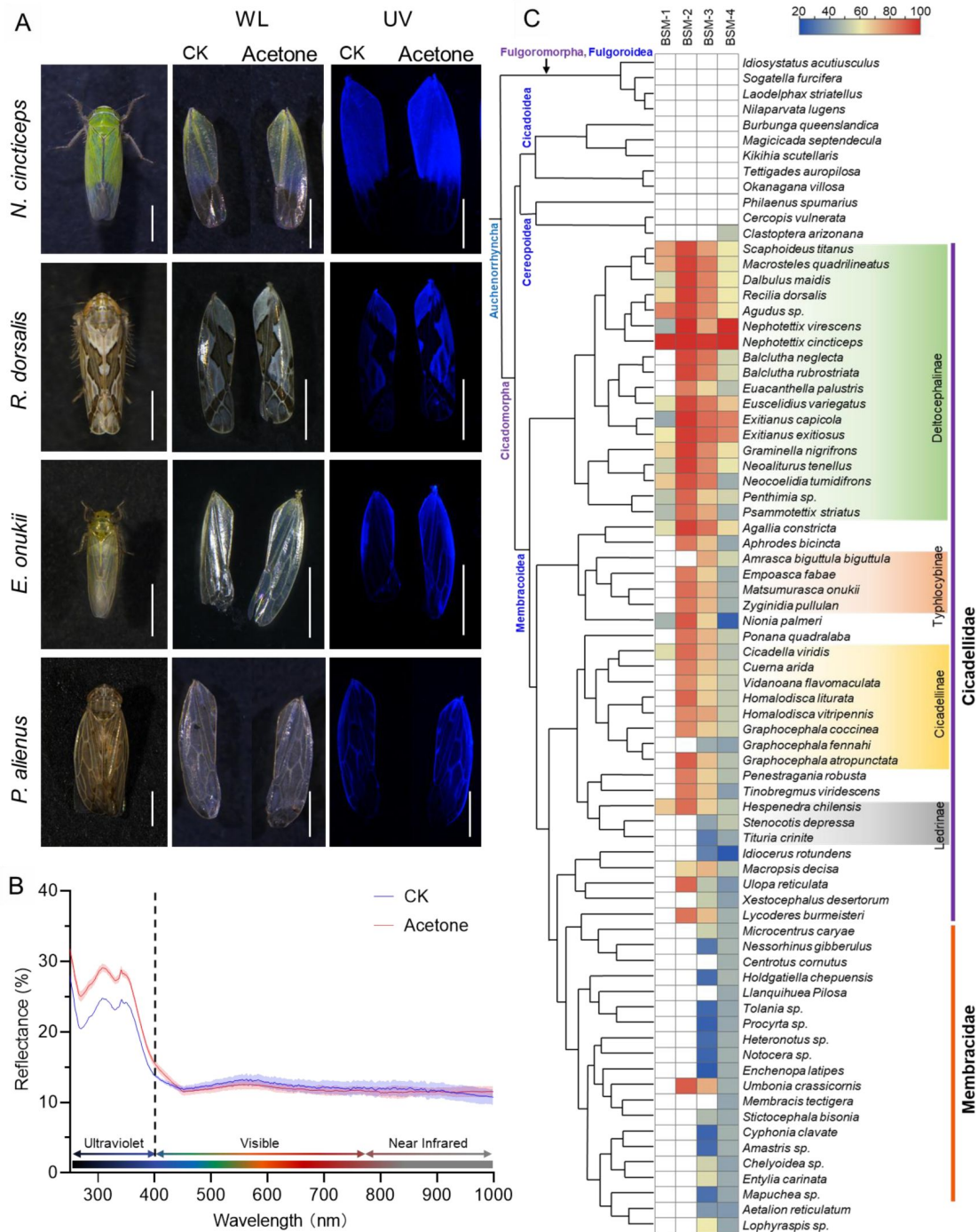


Figure 5.

Brochosome coating on Cicadellidae cuticle surface is essential for their optical qualities.

(A) Images of forewings of four leafhopper species before and after acetone treatment in white and ultraviolet light. Bar, 1mm. (B) Reflection spectra of the forewings of the *N. cincticeps* before and after acetone treatment. (C) Phylogenetic associations between the BSM and phylogeny of Hemipteran lineages. The left column describes the phylogeny of 76 representative species of Hemiptera with BSM proteins. In the right column, the presence of BSM-encoded genes in different Hemiptera species is illustrated along with their homology analysis with *N. cincticeps*. White indicates the absence of related genes; the color gradient represents the degree of nucleotide sequence similarity. All images are representative of at least three replicates.

the majority of Cicadellidae species with confirmed brochosome distribution, homologous gene distribution for BSM2 is detectable. In Cicadellidae species without brochosome distribution and in the majority of Membracidae, BSM2 genes are largely absent (**Figure 5C**). Since Membracidae evolved from Cicadellidae, we hypothesize that BSM homologous gene duplication/loss may be a key factor in brochosome formation/loss. In conclusion, our data suggest that in Cicadellidae, brochosome is synthesized by a conserved group of BSM coding genes, making it a widely distributed anti-reflective camouflage coating in Cicadellidae.

Discussion

Leafhoppers exhibit a distinctive grooming behavior, resulting in the deposition of a unique extra-cuticular coating known as brochosome (Rakitov, 2009). However, the precise characteristics and exact adaptive significance of this specialized coating remain largely unknown. Although they are widely believed to confer hydrophobicity, protecting leafhoppers from water and their own excreta (Rakitov & Gorb, 2013a), similar functionalities are attributed to cuticular waxes on the external surfaces of various insects, including leafhoppers (Andersen, 1979; Bello et al., 2022). Consequently, the functionalities of brochosomes may extend beyond this conventional understanding. In this study, we demonstrated that brochosomes exhibit robust anti-reflective properties, decreasing the reflectance of the leafhopper cuticle surface, particularly in the ultraviolet spectral range. This reduction may reduce visibility to visual predators, contributing to the evasion and predation avoidance (Vukusic & Sambles, 2003; Watson et al., 2017). Thus, brochosomes function as a natural camouflage coating on the leafhopper cuticle surface, playing a pivotal role in reducing leafhoppers' visibility to visual predators and offering a distinct advantage for their survival in predation scenarios as illustrated in **Fig. 6**.

The present work has elucidated the function of brochosome as a natural anti-reflective camouflage coating on leafhoppers' extra-cuticular, with its coverage directly influencing the anti-reflective properties of the cuticle surface (**Figure 2A-E** and **Figure 4A-G**). Previous studies suggested that arthropods employ various nanostructures formed by proteins, lipids, wax, or adhesive substances on their cuticle surfaces to achieve anti-reflectivity (Silberglied, 1979). These nanostructures, initially identified in the compound eyes of moths, have been subsequently found to be widely distributed on the cuticle surfaces of insects (Bernhard, 1965; Blagodatski et al., 2015; Chotard et al., 2022; Clapham & Hutley, 1973; Ho et al., 2016; Monteiro et al., 2019; Stavenga et al., 2005). The nano- and microstructures on insect cuticle surfaces generate a gradual change in refractive index near the surface, leading to antireflectivity (Raut et al., 2011). The degree of reflection reduction depends on the shape of these nanostructures, studies in *Drosophila melanogaster* have confirmed that modifying the morphology of nanostructures in compound eyes directly impacts their optical performance (Kryuchkov et al., 2020). In this study, a thorough examination of the structural protein composition of brochosome, coupled with RNAi-mediated suppression of BSM coding gene expression, yielded brochosomes with regular structure (dsGFP treatment) and structurally abnormal brochosomes (dsBSM treatment) (**Figure 4B-E**). Subsequent in vitro experiments provided further evidence of the correlation between the normal structure of brochosome and its anti-reflective performance (**Figure 4J** and **K**). This correlation is likely attributed to the fact that the diameter of the hollow pits in brochosome structure (approximately 100nm) is considerably smaller than most spectral wavelengths (Burks et al., 2022; Rakitov, 2009). As light passes through these small pits, it undergoes diffraction, while light passing through the ridges of brochosome induces scattering. The interference of diffracted and scattered light between different pits and ridges contributes to the observed extinction features in brochosome (Raut et al., 2011). Biomimetic material and mathematical modeling analyses have clarified the relationship between the extinction features of brochosome and the spacing between hollow pits (Banerjee et al., 2023; Lei et al., 2020; Wang et al., 2024; Yang et al., 2017). Therefore, the regular morphology of brochosome is a robust guarantee for its excellent anti-reflective performance.

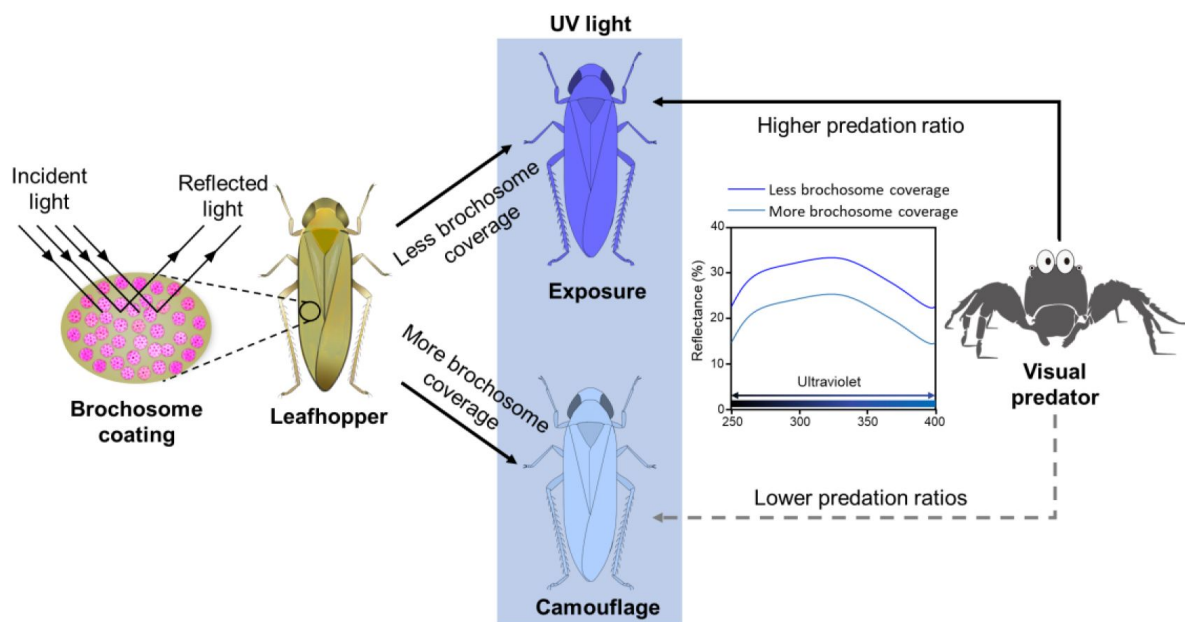


Fig. 6.

Brochosomes serve as an anti-reflective camouflage coating on the cuticle surface of leafhoppers.

Brochosomes effectively reduce the reflection of various wavelengths of light, particularly in the UV region. UV light is a crucial visual cue for numerous visual predators to identify and locate prey. Brochosomes efficiently decrease the UV light reflection on the surface of leafhoppers, thereby reducing their exposure risk to visual predators and facilitating evasion of visual predation.

Moreover, we noticed a distinct contrast in the anti-reflective capabilities when applying brochosomes with regular and irregular structures onto different substrates (quartz glass and brown planthopper wings), especially in the UV region (**Figure 4J** and **K**). This indicates that the reduction in reflected light is not only related to the structure of brochosome but also to other factors. SEM observations revealed that some brochosome is embedded in the grooves of the leafhopper cuticle surface or brown planthopper wings, which is not observed on the smooth surface of quartz glass (Figure 4B-figure supplement 6). When the wavelength of light is higher than the size of the structure and the surface has a gradient refractive index, light interacts entirely with the rough surface, causing the light to gradually bend (Raut et al., 2011). Combined with the complex structure formed by brochosomes and nanostructures on the cuticle surface, this result in multiple internal reflections of light within these structures, increasing light absorption and significantly enhancing anti-reflective performance. Furthermore, the aromatic ring structures, conjugated double bonds, and peptide bonds in the proteins of both the cuticle and brochosomes may contribute to the absorption of ultraviolet light, providing another potential reason for their anti-reflective capabilities in the ultraviolet spectral region. Therefore, the anti-reflective performance of brochosome on the insect cuticle surface might be a result of the combined effect of brochosome and cuticular nanostructures.

UV light serves as a crucial visual cue for various insect predators, enhancing foraging, navigation, mating behavior, and prey identification (Cronin & Bok, 2016; Morehouse et al., 2017; Silberglied, 1979). Predators such as birds, reptiles, and predatory arthropods often rely on UV vision to detect prey (Church et al., 1998; Li & Lim, 2005; Zou et al., 2011). However, UV reflectance from insect cuticles can disrupt camouflage, increasing the risk of detection and predation, as natural backgrounds like leaves, bark, and soil typically reflect minimal UV light (Endler, 1997; Li & Lim, 2005; Tovee, 1995). To mitigate this risk, insects often possess anti-reflective cuticular structures that reduce UV and broad-spectrum light reflectance. This strategy is widespread among insects, including cicadas, dragonflies, and butterflies, and has been shown to decrease predator detection rates (Hooper et al., 2006; Siddique et al., 2015; Zhang et al., 2006). For example, the compound eyes of moths feature hexagonal protuberances that reduce UV reflectance, aiding nocturnal concealment (Blagodatski et al., 2015; Stavenga et al., 2005). In butterflies, UV reflectance from eyespots on wings can attract predators, but reducing UV reflectance or eyespot size can lower predation risk and enhance camouflage (Chan et al., 2019; Lyytinen et al., 2004). Hence, the reflection of ultraviolet light from the insect cuticle surface increases the risk of predation by disrupting camouflage (Tovee, 1995). In this study, we utilized the jumping spider *P. paykulli* to explore the impact of the brochosome coating on the leafhopper's camouflage against predation. Hunting spiders, such as jumping spiders and wolf spiders, are primary predators of leafhoppers (Liu et al., 2015; Orazé & Grigarick, 1989) and the wavelength of UV light plays a crucial role as a visual cue for jumping spiders in identifying and locating their prey compare to the other wavelengths (Li & Lim, 2005; Zou et al., 2011). Our results revealed that *P. paykulli* displayed a preference for preying on *N. cincticeps* with a lower brochosome coverage on their cuticle surface (**Figure 2G-J** and **Figure 4H-I**). Thus, we hypothesize that the combination of brochosome and leafhopper cuticular microstructures exhibits superior anti-reflective performance in the UV region, effectively reducing UV reflection from leafhopper bodies and consequently diminishing the predator's ability to identify them. Additionally, spectral measurement data indicated that the wing reflection spectrum of the brown planthopper (**Figure 4J**) closely resembled the body spectrum of 25-day-old female *N. cincticeps* (**Figure 2E**). It is possible that, in the same living environment, other Hemipteran insects would be more conspicuous to visual predators than leafhoppers. Brochosome coating could provide an advantage for leafhoppers to evade visual predators, thereby enhancing their survival rates in interspecific competition. This adaptation could potentially favor the survival of Cicadellidae in environments with high predation pressure, allowing them to thrive in a wider range of environments. Additionally, within the leafhopper population, the brochosome coverage on the cuticle surface of older individuals was notably lower than that of younger individuals, making elderly leafhoppers more susceptible to be detected and captured by predators (Figure 2F-I-figure

supplement 1). This may help to eliminate older individuals who have lost their reproductive capacity in the population, thereby ensuring the vitality and reproductive power of the population. Therefore, we consider brochosome, as a form of anti-reflective camouflage coating on the leafhopper's cuticle surface, represents a more advanced evolutionary strategy against visual predation.

We successfully identified four proteins essential for the structure of brochosomes for the first time (**Figure 3**). These BSM-encoded genes exhibit typical orphan gene characteristics, consistent with previous findings (Li et al., 2022; Rakitov et al., 2018; Yuan et al., 2023). Orphan genes originate from processes like de novo evolution, horizontal gene transfer (HGT), and duplication–divergence (Light et al., 2014; Tautz & Domazet-Lošo, 2011). To elucidate the evolutionary origin of the four identified BSM genes, we aligned them with all prokaryotic genomes in NCBI and approximately 1000 arthropod genomes. Interestingly, the homologous genes of these BSM genes were exclusively found in leafhopper genomes in the family Cicadellidae, suggesting that the origin of these BSM genes might not be due to de novo evolution or horizontal gene transfer. Combining the species distribution of the four identified BSM genes (as well as their homology) (**Figure 5C**), we hypothesize that the BSMs were potentially originated from a process involving duplication–divergence. In this scenario, a new gene would be created through gene duplication or transposition, undergoing fast adaptive evolution and losing similarity to the original gene (Light et al., 2014; Tautz & Domazet-Lošo, 2011). For instance, BSM4 is widely present in Membracoidea species and also has homologous genes in *Clastoptera arizonana* of Cereopoidea, indicating that BSM4 may have a common ancestral gene in Cicadomorpha. The specialization of the distal segment of the Malpighian tubule is a characteristic shared by the three major lineages of the infraorder Cicadomorpha: Cercopoidea (spittle-bugs), Cicadoidea (cicadas), and Membracoidea (leafhoppers and treehoppers) (Rakitov, 2002). Cicadas coat the surface of their burrows with secretions from the Malpighian tubules, spittlebug nymphs have their integuments covered with foam synthesized by the Malpighian tubules, and leafhoppers coat their integuments with brochosomes synthesized by the Malpighian tubules (Chang et al., 2019; Rakitov, 2002). The different demands for secretions from the Malpighian tubules among these insects may be the main reason for the rapid adaptive evolution of BSM genes.

Among Auchenorrhyncha insects, most Cicadellidae (leafhoppers) secrete brochosomes, while many Membracidae (treehoppers) exhibit grooming behavior but lack specialized setae (Rakitov, 1996). Considering that Membracidae are derived from Cicadellidae (Dietrich et al., 2001), the formation of the symbiotic relationship between treehoppers and ants might be due to the loss of brochosome synthesis capability. Ants create a protective environment around treehoppers, deterring threats and reducing predation pressure, providing a relatively secure living environment, leading to lower predation pressure compared to leafhoppers (Del-Claro & Oliveira, 2000; Nelson & Mooney, 2022). Consequently, the need for the protective disguise represented by brochosomes is likely diminished. We hypothesize that environmental changes drive brochosome evolution, supported by the distribution of BSM2 and BSM3 paralogs in Auchenorrhyncha. BSM3 homologs are widespread in Auchenorrhyncha, while BSM2 is restricted to brochosome-secreting Cicadellidae and absent in most Membracidae. The high structural homology between BSM2 and BSM3, despite low sequence similarity, suggests that BSM2 arose from BSM3 duplication. We propose that gene duplication/loss events are the primary factors underlying differences in brochosome synthesis between Cicadellidae and Membracidae. Gene duplication/loss is a common mechanism for functional gain/loss in eukaryotes, promoting protein diversity and redundancy (Wong & Belov, 2012). Thus, gene duplication may be a key driver of brochosome formation and functional diversity in leafhoppers. In summary, our research has underscored the indispensable role of brochosomes as an essential anti-reflective camouflage coating for leafhoppers, conferring a significant advantage in evading visually-oriented predators. We have identified four key BSM encoding genes, a groundbreaking discovery in brochosome research, which suggests that gene duplication may be involved in the formation of these adaptive

structures. Our study not only deepens our understanding of leafhoppers' anti-predator strategies but also sheds light on the distinct evolutionary trajectories employed by these insects within their ecological niche. This contributes to the development of biomimicry and the advancement of state-of-the-art camouflage technologies. Furthermore, while our study has provided valuable insights into the structure and function of brochosomes, there are still gaps in our knowledge regarding the evolutionary origins of BSMs. Future research should aim to fill these gaps, as a more comprehensive understanding of BSMs and their role in brochosome formation will enhance our appreciation of the complex evolutionary processes that have shaped these unique structures. Additionally, the analysis of BSMs in a broader range of leafhopper species will help elucidate the relationship between BSM composition and the morphological and functional diversity of brochosomes, further informing our understanding of the ecological and evolutionary dynamics of this insect group.

Materials and methods

Insect

The leafhopper *N. cincticeps* and *R. dorsalis* adults were collected from a rice field in Jiaxing, Zhejiang Province, China in September 2020. *P. alienus* adults were collected from a wheat field in Shenyang, Liaoning Province, China in May 2020. *E. onukii* adult were collected from a tea-garden in Fuzhou, Fujian province, China in May 2020. The identity of the leafhopper species was confirmed using stereomicroscopy and mitochondrial cytochrome oxidase subunit 1 (CO1) sequence analysis (Supplementary Table 1). Collected leafhopper *N. cincticeps* and *R. dorsalis* are maintained in insect-proof greenhouses at $26 \pm 1^\circ\text{C}$ under a 16:8 h light:dark photoperiod and $50 \pm 5\%$ relative humidity on rice variety TaiChung Native 1(TN1).

The jumping spider *P. paykulli* were collected from a rice field in Ningbo, Zhejiang Province, China. Species identification was confirmed by assessing its morphological characteristics using stereomicroscopy and CO1 sequence analysis (Supplementary Table 1). Jumping spiders were fed about their own mass of leafhoppers *N. cincticeps* three times each week and examined 14-52 days following collection, as previously reported (Taylor et al., 2014 [DOI](#)).

Electron microscopy

The Malpighian tubules of *N. cincticeps* were dissected using an astereomicroscope, serially fixed overnight at 4°C with 2.5% glutaraldehyde in 0.1 mol/L PBS, and post-fixed for 2 hours at room temperature with 1% OsO₄. Following that, samples were dehydrated in a series of ethanol solutions (50%, 70%, 80%, 90%, and 95%), then permeabilized with 100% ethanol and 100% acetone, followed by a series of epoxies in acetone (50%, 75%, and 100%). The pierced tissue was then immersed in epoxy resin and baked for more than 24 hours at 70°C . Ultrathin slices were cut and stained with uranyl acetate for 15 minutes and lead citrate for 5 minutes, and examined by Hitachi H7800 microscope (Hitachi, Japan).

Scanning electron microscopy was used to observe the morphology of leafhopper brochosomes. Samples were prepared as previously described (Rakitov & Gorb, 2013a [DOI](#)). Leafhopper forewings were obtained by dissection under the stereomicroscope and glued onto SEM metal stubs with a double-sided carbon tape. The dried samples were coated with gold for 1 min in a MC1000 Ion Sputter Coater (Hitachi, Japan) and viewed by a HITACHI Regulus 8100 SEM (Hitachi, Japan).

Spider predation experiment

Jumping spiders were starved for three days prior to the predation experiment to ensure that participating spiders were hungry enough to attack leafhoppers in our predation preference tests, but not so hungry that they would attack the first prey they encountered (without being choosy)

(Taylor et al., 2014 [DOI](#)). For the jumping spiders to prey on, we placed two leafhoppers in a 9 cm diameter test arena (the bottom of the dish was covered with filter paper). Before starting the test, the spiders were placed in the arena for 15 minutes to adapt, and then they were released and permitted to prey on the leafhoppers. We directly observed the feeding process of the jumping spiders and recorded when the spiders attacked the first leafhopper and which leafhopper they preyed on. The predation finishes when the first leafhopper is attacked or when the jumping spider does not attack any leafhopper within 5 minutes. Each pair of jumping spider predation tests were independently replicated more than 60 times.

The distribution of cuticle surface brochosomes and the optical properties of male and female insects at various periods post-eclosion

To investigate the distribution of brochosomes on the cuticle surface of adult *N. cincticeps* at various post-eclosion periods, we first collected female and male adult leafhoppers at 5-, 10-, 15-, 20-, and 25-days post-eclosion and dissected the forewings under a Nikon SMZ25 microscope (Nikon, Japan). The forewings were gold-sprayed and examined using HITACHI Regulus 8100 SEM (Hitachi, Japan). At each time point, 30 samples were randomly selected for SEM imaging. A $10\ \mu\text{m} \times 10\ \mu\text{m}$ area within the SEM images was used to assess brochosome coverage, and ImageJ was employed for analysis (Schneider et al., 2012 [DOI](#)). Since brochosomes are synthesized at the distal section of the Malpighian tubule and the morphology of the Malpighian tubule in leafhoppers can also reflect brochosomes synthesis, we used Nikon SMZ25 microscope (Nikon, Japan) to examine the morphology of the Malpighian tubule in female and male adult leafhoppers at 5-, 10-, 15-, 20-, and 25-days post-eclosion.

To examine the optical properties of male and female insects at various periods post-eclosion, we first collected female and male adult leafhoppers at 5-, 10-, 15-, 20-, and 25-days post-eclosion, freeze-killed them and placed them under white or UV light for observation through a Nikon SMZ25 microscope (Nikon, Japan). Following that, the forewings of the leafhoppers were meticulously dissected, and the specular reflectance spectra of the forewings ($25 \times 25\ \mu\text{m}$) were precisely measured utilizing a 20/30 PV UV-Vis-NIR microspectrophotometer (CRAIC Technologies Inc., USA). UV-Vis-NIR reflectance spectra (250 to 1000 nm) were obtained using a $10\times$ UV-absorbing glass objective lens, with a minimum of five replicates per sample.

Identification of brochosome structural protein-coding genes

Candidate brochosome structural protein-coding genes were identified through a combined analysis of transcriptomic and proteomic data. To knock down the expression of these candidates, RNAi was conducted by microinjecting double-stranded RNA (dsRNA) into the abdomens of fifth instar nymphs. The dsRNAs targeting 500-1000 bp regions of Malpighian tubule-specific expression genes or GFP were synthesized in vitro using the T7 RiboMAX Express RNAi System (Promega, USA). The 50 fifth instar nymphs of *N. cincticeps* were microinjected with 30 nl dsRNA ($0.5\ \mu\text{g}\ \mu\text{l}^{-1}$) using the Nanoject II Auto-Nanoliter Injector (Drummond, USA). Thereafter, they were transferred to healthy rice seedlings for recovery. Forewings of treated leafhoppers were collected 7 days after microinjection (about 4-5 days post-eclosion), sprayed with gold, and analyzed by HITACHI Regulus 8100 SEM (Hitachi, Japan). Screening identified four brochosome structural protein-coding genes, and inhibiting their expression resulted in significant changes in brochosome morphology. The full-length sequence of these genes was obtained using by 5'-rapid amplification of cDNA ends (RACE) and 3'-RACE using SMARTer RACE 5'/3' Kit (Clontech, USA).

To analyze the temporal expression profiles of four brochosome structural protein-coding genes (GenBank accession number PP273097, PP273098, PP273099, PP273100), the whole body, salivary gland, midgut, ovary, and testis were dissected from 200 adult insects, total RNA in different tissue were extracted using TRIzol Reagent (Invitrogen, USA). The relative expression of these genes in

different tissues of *N. cincticeps* adults was detected by RT-qPCR assay using a QuantStudio™ 5 Real-Time PCR system (Thermo Fisher Scientific, USA). The detected transcript levels were normalized to the transcript level of the housekeeping gene elongation factor 1 alpha (EF1α) (GenBank accession number AB836665) and estimated by the $2^{-\Delta\Delta C_t}$ (cycle threshold) method. Using the same procedure, the expression levels of these brochosome structural protein-coding genes were evaluated in female and male adult leafhoppers at 5-, 10-, 15-, 20-, and 25-days post-eclosion.

To investigate the effects of four BSM-encoding genes on brochosome morphology and distribution, dsRNAs were designed to target two non-overlapping regions within the sequences of these genes. These dsRNAs were microinjected into the abdomens of leafhoppers both individually and in combination, with dsRNA targeting GFP serving as a negative control. Following a seven-day incubation period post-injection, changes in BSM gene expression were assessed in both control and treated groups via RT-qPCR. The morphology and distribution of brochosomes on the leafhopper cuticle were examined and documented using SEM. Additionally, ImageJ software was employed to quantify the proportion of morphologically abnormal brochosomes and the distribution area of brochosomes on the cuticle surface after injection of the dsRNA mixtures targeting the two non-overlapping regions of each BSM gene, as well as after dsGFP treatment.

Knocking down in vivo expression of BSM-coding genes in *N. cincticeps*

Following the prior procedure, dsRNA produced from four BSM-coding genes was injected into the abdomen of fifth instar nymphs of *N. cincticeps* by microinjection, and a control group injected with dsGFP was established. After a seven-day incubation post-injection, alterations in the expression of the four BSM genes were monitored in the dsBSM and dsGFP treated leafhoppers using RT-qPCR. The synthesis of brochosomes within the Malpighian tubules was investigated through TEM analysis. The morphology and distribution of brochosomes on the leafhopper cuticle were observed and imaged using SEM. To examine the optical properties, the dsBSM and dsGFP treated leafhoppers, freeze-killed them and placed them under white or UV light for observation through a Nikon SMZ25 microscope (Nikon, Japan). Subsequently, the forewings of the leafhoppers were meticulously dissected, and the specular reflectance spectra of the forewings ($25 \times 25 \mu\text{m}$) were precisely measured utilizing a 20/30 PV UV-Vis-NIR microspectrophotometer (CRAIC Technologies Inc., USA). UV-Vis-NIR reflectance spectra (250 to 1000 nm) were obtained using a $10\times$ UV-absorbing glass objective lens, with a minimum of five replicates per sample. Finally, the predation preference of the jumping spiders on the dsBSM and dsGFP treated leafhoppers was examined by a predation experiment.

In vitro measurements were conducted to assess the relationship between brochosome morphology and their optical performance

To collect approximately 500 *N. cincticeps* treated with dsGFP or dsBSM, dissect the forewings of the leafhoppers and immerse them in acetone in a 50 mL centrifuge tube. Place the centrifuge tubes on a room temperature orbital shaker at 50 rpm for 12 hours to separate the forewings from the leafhoppers due to friction. Following this, centrifuge at 1000g for 10 minutes to separate the brochosomes. Resuspend the particles in fresh acetone, briefly sonicate, then centrifuge at 1000g for 10 minutes. Repeat this process three times. Resuspend the collected brochosomes from dsGFP and dsBSM-treated leafhopper forewings in acetone and adjust the two suspension solutions to the same OD280 reading. Next, using a pipette, carefully drop the brochosome solution onto quartz slides and brown planthopper wings. Allow the acetone to completely evaporate at room temperature. At the same time, we also set up quartz slides and brown planthopper wings that were treated with acetone alone as negative controls. Prepared quartz slides and brown planthopper wings can be observed under SEM to examine the distribution of brochosomes.

Additionally, UV-Vis-NIR reflectance spectra (250 to 1000 nm) were obtained using a 20/30 PV UV-Vis-NIR microspectrophotometer (CRAIC Technologies Inc., USA), with a minimum of five replicates per sample.

To investigate the correlation between BSM proteins and the optical characteristics of brochosomes, the four genes encoding BSM proteins were separately cloned into pET-28a or pGEX-4T2 vectors. Following induction, the expressed proteins were purified from cell lysates via batch affinity chromatography using either glutathione (GSH) agarose or nickel-nitrilotriacetic acid (Ni-NTA) agarose, depending on the fusion tag. GST protein was produced following the identical protocol from cells harboring the empty pGEX-4T2 vector. The purified BSM fusion proteins and GST protein were normalized to an equivalent OD₂₈₀ value. These protein solutions were then meticulously spotted onto quartz slides and permitted to dry completely at ambient temperature. In parallel, quartz slides treated exclusively with PBS served as negative controls. The quartz slides were subsequently analyzed to acquire UV-Vis-NIR reflectance spectra (ranging from 250 to 1000 nm) using a 20/30 PV UV-Vis-NIR microspectrophotometer (CRAIC Technologies Inc., USA), with a minimum of five technical replicates conducted per sample.

Removal of leafhopper forewing brochosome by acetone

Previous research has shown that acetone may efficiently remove brochosomes from the cuticle surface of leafhoppers (Rakitov et al., 2018). We manually removed the brochosomes from the forewings of leafhoppers using acetone and examined the imaging under white or UV light for observation through a Nikon SMZ25 microscope (Nikon, Japan). UV-Vis-NIR reflectance spectra (250 to 1000 nm) were obtained using a 20/30 PV UV-Vis-NIR microspectrophotometer (CRAIC Technologies Inc., USA), with a minimum of five replicates per sample.

Bioinformatics analysis and phylogenetic tree

The brochosome is a unique secretion produced by leafhoppers (Cicadellidae). In order to determine the distribution of brochosome structural protein-coding genes in Hemiptera, the brochosome structural protein-coding genes were used as queries to search for homologous sequences in Hemiptera transcriptomes. We examined 116 Hemiptera species, encompassing major families (Acanaloniidae, Achilidae, Aetalionidae, Aphrophoridae, Caliscelidae, Cercopidae, Cicadellidae, Cicadidae, Cixiidae, Clastopteridae, Delphacidae, Derbidae, Dictyopharidae, Epipygidae, Eurybrachidae, Flatidae, Fulgoridae, Issidae, Machaerotidae, Melizoderidae, Membracidae, Myerslopiidae, Nogodinidae, Peloridiidae, Ricaniidae, Tettigarctidae, Tettigometridae, Theaceae, Tropiduchidae). Transcriptome data of 116 Hemiptera species were downloaded from the NCBI Sequence Read Archive (Supplementary Table 2), and assembled using SOAPdenovo-Trans (version 1.01) (Xie et al., 2014). The resulting transcripts were filtered to remove potential contaminants. We compared four identified BSM-encoded genes with these transcriptomes using the tblastn (E-value < 1.0e-5). Heatmaps were generated using TBtools software (Chen et al., 2023). At least one homologous gene for the four BSM genes was identified in 66 species. A species phylogenetic tree for these 66 Hemiptera species was constructed following the methods outlined in previous studies (Johnson et al., 2018). OrthoFinder to identify orthogroups and infer the species tree, obtaining 1683 single-copy orthologous genes after filtering (Emms & Kelly, 2019). MAFFT was employed for sequence alignment, followed by trimming with trimAI (Capella-Gutiérrez et al., 2009; Katoh & Standley, 2013). The best amino acid substitution model was selected using ModelFinder based on the Bayesian Information Criterion (BIC) (Kalyaanamoorthy et al., 2017). The maximum likelihood phylogenetic tree was constructed using IQ-TREE v1.6.8 with 1000 ultrafast bootstrap replicates (Minh et al., 2020). All analyses were conducted using PhyloSuite v1.2.2, which integrates MAFFT, trimAI, ModelFinder, and IQ-TREE into a unified pipeline (Capella-Gutiérrez et al., 2009; Emms & Kelly, 2019; Kalyaanamoorthy et al., 2017; Katoh & Standley, 2013; Minh et al., 2020; Zhang et al., 2020).

The amino acid sequences of BSM2 and BSM3 were aligned using the Clustal W multiple sequence alignment program (Larkin et al., 2007 [↗](#)). AlphaFold2 was employed to predict the three-dimensional structures of BSM2 and BSM3, and structural alignments were conducted using PyMOL (DeLano, 2002 [↗](#); Jumper et al., 2021 [↗](#)). In Membracoidea, the maximum likelihood (ML) phylogenetic tree for BSM1-4 was constructed using MAFFT, trimAI, ModelFinder, and IQ-TREE in PhyloSuite (Capella-Gutiérrez et al., 2009 [↗](#); Emms & Kelly, 2019 [↗](#); Kalyaanamoorthy et al., 2017 [↗](#); Katoh & Standley, 2013 [↗](#); Minh et al., 2020 [↗](#); Xiang et al., 2023 [↗](#); Zhang et al., 2020 [↗](#)).

Statistical analyses

All experiments were performed at least three independent replicates and statistical analyses were performed with GraphPad Prism8.0 software. Statistical significance was calculated by a two-tailed Student's t-test, one-way ANOVA, two-way ANOVA and/or unpaired Student's t-test. P-values < 0.05 were considered statistically significant.

Data availability

All the data needed to understand and assess the conclusions of this research are available in the article, Supplementary Materials, and GenBank (accession number PP273097, PP273098, PP273099, PP273100). Any additional information required to reanalyse the data reported in this paper is available from the lead contact upon request.

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Additional information

Author contributions

W.W. J.-M.L., C.-X.Z., and J.-P.C. designed research; W.W. Q.M. Z.Y. Z.L. performed research; W.W. and Z.Y. analyzed data; and W.W. H.-W.S., J.-M.L., C.-X.Z., and J.-P.C. wrote the paper. All authors gave final approval for publication. All authors have read and agreed to the published version of the manuscript.

Additional files

Supplemental Figures and Tables [↗](#)

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Reviewer #1 (Public review):

Summary:

Evading predation is of utmost importance for most animals and camouflage is one of the predominant mechanisms. Wu et al. set out to test the hypothesis of a unique camouflage system in leafhoppers. These animals coat themselves with brochosomes, which are spherical nanostructures that are produced in the Malpighian tubules and are distributed on the cuticle after eclosion. Based on previous findings on reflectivity properties of brochosomes, the authors provide convincing evidence that these nanostructures indeed reduce reflectivity of the animals thereby reducing predation by jumping spiders. Further, they identify four proteins, which are essential for proper development and function of brochosomes: In RNAi experiments, the regular brochosome structure is lost, the reflectivity reduced and the respective animals are prone to increased predation. Finally, the authors provide phylogenetic sequence analyses and speculate about the evolution of these genes.

Strengths:

The study is very comprehensive including careful optical measurements, EM and TM analysis of the nanoparticles and their production line in the malpighian tubules, in vivo predation tests and knock-down experiments to identify essential proteins. Indeed, the results are very convincingly in line with the starting hypothesis such that the study robustly assigns a new biological function to the brochosome coating system.

A key strength of the study is that the biological relevance of the brochosome coating is convincingly shown by an in vivo predation test using a known predator from the same habitat.

Another major step forward is an RNAi screen, which identified four proteins, which are essential for the brochosome structure (BSMs). After respective RNAi knock-downs, the brochosomes show curious malformations that are interesting in terms of the self-assembly of these nanostructures. The optical and in vivo predation tests provide excellent support for the model that the RNAi knock-down leads to a change of brochosomes structure, which reduces reflectivity, which in turn leads to a decrease of the antipredatory effect.

Conclusion:

The authors successfully tested their hypothesis in a multidisciplinary approach and convincingly assigned a new biological function to the brochosomes system. The results fully support their claims on the involvement of the four BSM genes in brochosome structure, the relevance of brochosomes for predation avoidance and they provide evidence for the evolution of these genes.

The work is a very interesting study case of the evolutionary emergence of a new system to evade predators. Based on this study, the function of the BSM genes could now be studied in other species to provide insights into putative ancestral functions. Further, studying the self-

assembly of such highly regular complex nano-structures will be strongly fostered by the identification of the four key structural genes.

<https://doi.org/10.7554/eLife.99639.2.sa2>

Reviewer #2 (Public review):

Summary:

In this manuscript, the authors investigate the optical properties of brochosomes produced by leafhoppers. They hypothesize that brochosomes reduce light reflection on the leafhopper's body surface, aiding in predator avoidance. Their hypothesis is supported by experiments involving jumping spiders. Additionally, the authors employ a variety of techniques including micro-UV-Vis spectroscopy, electron microscopy, transcriptome and proteome analysis, and bioassays. This study is highly interesting, and the experimental data is well-organized and logically presented.

Strengths:

The use of brochosomes as a camouflage coating has been hypothesized since 1936 (R.B. Swain, *Entomol. News* 47, 264-266, 1936) with evidence demonstrated by similar synthetic brochosome systems in a number of recent studies (S. Yang, et al. *Nat. Commun.* 8:1285, 2017; L. Wang, et al., *PNAS*. 121: e2312700121, 2024). However, direct biological evidence or relevant field studies have been lacking to directly support the hypothesis that brochosomes are used for camouflage. This work provides the first biological evidence demonstrating that natural brochosomes can be used as a camouflage coating to reduce the leafhoppers' observability to their predators. The design of the experiments is novel.

Weaknesses:

- (1) The observation that brochosome coatings become sparse after 25 days in both male and female leafhoppers, resulting in increased predation by jumping spiders, is intriguing. However, since leafhoppers consistently secrete and groom brochosomes, it would be beneficial to explore why brochosomes become significantly less dense after 25 days.
- (2) The authors demonstrate that brochosome coatings reduce UV (specular) reflection compared to surfaces without brochosomes, which can be attributed to the rough geometry of brochosomes as discussed in the literature. However, it would be valuable to investigate whether the proteins forming the brochosomes are also UV absorbing.
- (3) The experiments with jumping spiders show that brochosomes help leafhoppers avoid predators to some extent. It would be beneficial for the authors to elaborate on the exact mechanism behind this camouflage effect. Specifically, why does reduced UV reflection aid in predator avoidance? If predators are sensitive to UV light, how does the reduced UV reflectance specifically contribute to evasion?
- (4) An important reference regarding the moth-eye effect is missing. Please consider including the following paper: Clapham, P. B., and M. C. Hutley. "Reduction of lens reflection by the 'Moth Eye' principle." *Nature* 244: 281-282 (1973).
- (5) The introduction should be revised to accurately reflect the related contributions in literature. Specifically, the novelty of this work lies in the demonstration of the camouflage effect of brochosomes using jumping spiders, which is verified for the first time in leafhoppers. However, the proposed use of brochosome powder for camouflage was first described by R.B. Swain (R.B. Swain, Notes on the oviposition and life history of the leafhopper *Oncometopta undata* Fabr. (Homoptera: Cicadellidae), *Entomol. News*. 47: 264-266

(1936)). Recently, the antireflective and potential camouflage functions of brochosomes were further studied by Yang et al. based on synthetic brochosomes and simulated vision techniques (S. Yang, et al. "Ultra-antireflective synthetic brochosomes." *Nature Communications* 8: 1285 (2017)). Later, Lei et al. demonstrated the antireflective properties of natural brochosomes in 2020 (C.-W. Lei, et al., "Leafhopper wing-inspired broadband omnidirectional antireflective embroidered ball-like structure arrays using a nonlithography-based methodology." *Langmuir* 36: 5296-5302 (2020)). Very recently, Wang et al. successfully fabricated synthetic brochosomes with precise geometry akin to those natural ones, and further elucidated the antireflective mechanisms based on the brochosome geometry and their role in reducing the observability of leafhoppers to their predators (L. Wang et al. "Geometric design of antireflective leafhopper brochosomes." *Proceedings of the National Academy of Sciences* 121: e2312700121 (2024)).

Comments on revisions:

In this revision, the authors have addressed some of the key concerns I raised in our previous comments. However, a few issues remain unaddressed. Additionally, the new experimental data introduced in the manuscript require further clarification, which I outline below.

(1) As I pointed out in my previous review comments, "The use of brochosomes as a camouflage coating has been hypothesized since 1936 (R.B. Swain, *Entomol. News* 47, 264-266, 1936) with evidence demonstrated by similar synthetic brochosome systems in a number of recent studies (S. Yang, et al. *Nat. Commun.* 8:1285, 2017; L. Wang, et al., *PNAS*. 121: e2312700121, 2024). However, direct biological evidence or relevant field studies have been lacking to directly support the hypothesis that brochosomes are used for camouflage." While the authors did cite the original hypothesis proposed by R.B. Swain (1936), they have omitted important references that provide evidence on the use of antireflective properties of brochosomes for camouflage in a synthetic setting (see for example, Fig. 5a of S. Yang, et al. *Nat. Commun.* 8:1285, 2017). The authors are recommended to revise the Abstract and Introduction accordingly to ensure a fair and accurate representation of the existing literature.

(2) The antireflection mechanisms of brochosome structures have been discussed in detail, specifically, how their geometries (i.e., brochosome diameter and pore size) contribute to reducing UV reflectance (L. Wang, et al., *PNAS*. 121: e2312700121, 2024 and P. Banerjee, et al., *Advanced Photonics Research* 4:2200343, 2023). The authors should incorporate these recent findings into their discussion (line 381 - line 383 of the manuscript).

(3) The authors presented new data brochosomes deposited on a quartz slide and measured their reflectance across UV, visible light, and infrared wavelengths. Since reflectance is highly sensitive to the uniformity of brochosome coverage on the substrate, it is crucial to quantify this coverage across the measurement area for comparison. While the authors include SEM images to illustrate the packing of brochosomes on both the leafhopper wing and the quartz substrate (Fig. S7) at a microscopic scale (~10 μm view), it would be beneficial to also provide SEM images at a larger scale (e.g., 100 μm - 1 mm) and quantify the density of brochosomes per unit area for comparison.

(4) For the negative control using acetone to remove the brochosomes the leafhopper wing, have the authors confirmed the absence of brochosomes after treatment? If so, the authors should explicitly indicate this for clarity.

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Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public review):*Summary:*

Evading predation is of utmost importance for most animals and camouflage is one of the predominant mechanisms. Wu et al. set out to test the hypothesis of a unique camouflage system in leafhoppers. These animals coat themselves with brochosomes, which are spherical nanostructures that are produced in the Malpighian tubules and are distributed on the cuticle after eclosion. Based on previous findings on the reflectivity properties of brochosomes, the authors provide very good evidence that these nanostructures indeed reduce the reflectivity of the animals thereby reducing predation by jumping spiders. Further, they identify four proteins, which are essential for the proper development and function of brochosomes. In RNAi experiments, the regular brochosome structure is lost, the reflectivity reduced and the respective animals are prone to increased predation. Finally, the authors provide some phylogenetic sequence analyses and speculate about the evolution of these essential genes.

Strengths:

The study is very comprehensive including careful optical measurements, EM and TM analysis of the nanoparticles and their production line in the malpighian tubules, in vivo predation tests, and knock-down experiments to identify essential proteins. Indeed, the results are very convincingly in line with the starting hypothesis such that the study robustly assigns a new biological function to the brochosome coating system.

A key strength of the study is that the biological relevance of the brochosome coating is convincingly shown by an in vivo predation test using a known predator from the same habitat.

Another major step forward is an RNAi screen, which identified four proteins, which are essential for the brochosome structure (BSMs). After respective RNAi knock-downs, the brochosomes show curious malformations that are interesting in terms of the self-assembly of these nanostructures. The optical and in vivo predation tests provide excellent support for the model that the RNAi knock-down leads to a change of brochosomes structure, which reduces reflectivity, which in turn leads to a decrease of the antipredatory effect.

Thank you very much for your positive feedback and insightful comments on our manuscript. We are delighted that you acknowledge the efforts we have made in studying the components and functions of Brochosomal proteins. We have carefully considered your suggestions and have thoroughly revised the manuscript to address the shortcomings identified in our original submission. We hope that the revised version meets with your approval. Below, please find our detailed point-by-point responses.

Weaknesses:

The reduction of reflectivity by aberrant brochosomes or after ageing is only around 10%. This may seem little to have an effect in real life. On the other hand, the in vivo predation tests confirm an influence. Hence, this is not a real weakness of the study - just a note to reconsider the wording for describing the degree of reflectivity.

Thank you for your valuable suggestions. Based on your recommendations, we have revised the manuscript accordingly. Although the absolute reduction in light reflection due to Brochosomal coverage is approximately 10%, the relative decrease in light reflection on the leafhopper's surface is nearly 30%. Specifically, in the ultraviolet region, the reflection is reduced from about 30% to 20%, and in the visible light region, it is reduced from 20% to 10%. For detailed revisions, please refer to lines 151-156 of the revised manuscript.

The single gene knockdowns seemed to lead to a very low penetrance of malformed brochosomes (Figure Supplement 3). Judging from the overview slides, less than 1% of brochosomes may have been affected. A quantification of regular versus abnormal particles in both, wildtype and RNAi treatments would have helped to exclude that the shown aberrant brochosomes did not just reflect a putative level of "normal" background defects. Of note, the quadruple knock-down of all BSMs seemed to lead to a high penetrance (Figure 4), which was already reflected in the microtubule production line. While the data shown are convincing, a quantification might strengthen the argument.

While the RNAi effects seemed to be very specific to brochosomes and therefore very likely specific, an off-target control for RNAi was still missing. Finding the same/similar phenotype with a non-overlapping dsRNA fragment in one off-target experiment is usually considered required and sufficient. Further, the details of the targeted sequence will help future workers on the topic.

Thank you for your valuable suggestions. Based on your recommendations, we have synthesized dsRNA targeting two non-overlapping regions of the coding sequences for four Brochosomal structural protein genes. These dsRNAs were injected individually and in combination for each gene. Our RNAi experiments for each BSM gene demonstrated that both individual and combined injections significantly suppressed the expression of the target genes, with the combined injection yielding slightly better silencing efficiency. Statistical analysis of the SEM observations revealed that the combined injection of dsRNAs targeting two non-overlapping regions led to a 60-70% reduction in the surface area coverage of Brochosomes. Additionally, approximately 20% of the remaining Brochosomes exhibited significant morphological changes. For detailed revisions, please refer to lines 199-211 of the revised manuscript, as well as Figures 3A and 3C, and Supplementary Figures 4 and 5.

*The main weakness in the current manuscript may be the phylogenetic analysis and the model of how the genes evolved. Several aspects were not clearly or consistently stated such that I felt unsure about what the authors actually think. For instance: Are all the 4 BSMs related to each other or only BSM2 and 3? If so, not only BSM2 and 3 would be called "paralogs" but also the other BSMs. If they were all related, then a phylogenetic tree including all BSMs should be shown to visualize the relatedness (including the putative ancestral gene if that is the model of the authors). Actually, I was not sure about how the authors think about the emergence of the BSMs. Are they real orphan genes (i.e. not present outside the respective clade) or was there an ancestral gene that was duplicated and diverged to form the BSMs? Where in the phylogeny does the first of the BSMs or ancestral proteins emerge (is the gene found in *Clastoptera arizonana* the most ancestral one?)? Maybe, the evolution of the BSMs would have to be discussed individually for each gene as they show somewhat different patterns of emergence and loss (BSM4 present in all species, the others with different degrees of phylogenetic restriction).*

Thank you very much for your constructive feedback on our phylogenetic analysis and the modeling of gene evolution. We fully agree with your insights and acknowledge that the evolutionary analysis of BSM genes remains somewhat ambiguous. This ambiguity is

primarily due to the limited research on the precise structural protein composition of Brochosomes. While proteomics studies have analyzed and discussed the structural proteins of Brochosomes, the accurate composition of these proteins is still poorly understood. In this study, we identified four BSM proteins, but given the intricate structure of Brochosomes as proteinaceous spheres, we believe there may be additional BSM genes that have not yet been identified. Moreover, despite the presence of over ten thousand species within the Cicadomorpha, only three species have genome sequences available, and fewer than a hundred species have transcriptome sequencing data. The scarcity of research on Brochosomes, as well as the limited availability of genomic and transcriptomic data, poses significant challenges for our phylogenetic analysis and understanding of BSM gene evolution.

Based on your suggestions, we have revised the manuscript accordingly. Specifically, we have updated Figure 5C by including ten additional species from Cereopoidea, Cicadoidea, and Fulgoroidea to better illustrate that BSM genes are true orphan genes. We have also added a phylogenetic tree of BSM genes within Cicadidae in Supplementary Figure 3. Additionally, we have expanded the discussion of BSM gene evolution in the manuscript (lines 503-556). For detailed revisions, please refer to Figure 5C, Supplementary Figure 3, and lines 507-585 of the revised manuscript.

*Related to these questions I remained unsure about some details in Figure 5. On what kind of analysis is the phylogeny based? Why are some species not colored, although they are located on the same branch as colored ones? What is the measure for homology values - % identity/similarity? The homology labels for *Nephotetix cincticeps* and *N. virescens* seem to be flipped: the latter is displayed with 100% identity for all genes with all proteins while the former should actually show this. As a consequence of these uncertainties, I could not fully follow the respective discussion and model for gene evolution.*

Thank you very much for your insightful comments and suggestions. We have carefully considered your feedback and have thoroughly revised our manuscript accordingly. Specifically, we have enhanced the description of the phylogenetic analysis process to provide greater clarity and transparency, with the detailed methods now included in lines 789-798. Regarding Figure 5C, we appreciate your attention to the coloring scheme. We would like to clarify that the family Cicadellidae comprises 25 subfamilies, many of which are represented by only one species in our figure. To ensure clarity and meaningful representation, we have chosen to color only those subfamilies with more than three species, thereby avoiding visual clutter and emphasizing the most relevant taxonomic groups. Additionally, we have corrected the inverted homology labels for *Nephotetix cincticeps* and *Nephotetix virescens* to ensure the accuracy and consistency of our data presentation.

Conclusion:

The authors successfully tested their hypothesis in a multidisciplinary approach and convincingly assigned a new biological function to the brochosomes system. The results fully support their claims - only the quantification of the penetrance in the RNAi experiments would be helpful to strengthen the point. The author's analysis of the evolution of BSM genes remained a bit vague and I remained unsure about their respective conclusions.

The work is a very interesting study case of the evolutionary emergence of a new system to evade predators. Based on this study, the function of the BSM genes could now be studied in other species to provide insights into putative ancestral functions. Further, studying the self-assembly of such highly regular complex nano-structures will be strongly fostered by the identification of the four key structural genes.

Reviewer #1 (Recommendations for the authors):

Main manuscript:

Please consider the annotated pdf with suggestions for wording and comments at the authors' discretion:

Thank you very much for your detailed suggestions and comments provided in the annotated PDF. We have carefully reviewed each of your points and have revised the manuscript accordingly. All changes have been highlighted in red text for your convenience. The revised manuscript with tracked changes is available for your review. We believe these revisions have improved the clarity and quality of our manuscript. Thank you again for your valuable feedback.

Supplementary Figure 2 C:

Y-axes:

- label: "surface coverage in %"

- there are different scale values for the different days (e.g. 80-105 for day 5 and 0-80 at day 25). As a comparison between days is interesting, it would help to have the same scale values for all. That would show the decrease more intuitively.

Thank you very much for your suggestion regarding the Y-axis in Supplementary Figure 2C. We agree that using a consistent scale across all time points is essential for clear and intuitive comparison. In the revised manuscript, we have standardized the Y-axis scale for Supplementary Figure 2C to a uniform range of 0-100% for all days. This change allows for a more straightforward visualization of the decreasing trend in surface coverage over time.

Reviewer #3 (Public review):

Summary:

In this manuscript, the authors investigate the optical properties of brochosomes produced by leafhoppers. They hypothesize that brochosomes reduce light reflection on the leafhopper's body surface, aiding in predator avoidance. Their hypothesis is supported by experiments involving jumping spiders. Additionally, the authors employ a variety of techniques including micro-UV-Vis spectroscopy, electron microscopy, transcriptome and proteome analysis, and bioassays. This study is highly interesting, and the experimental data is well-organized and logically presented.

Strengths:

The use of brochosomes as a camouflage coating has been hypothesized since 1936 (R.B. Swain, Entomol. News 47, 264-266, 1936) with evidence demonstrated by similar synthetic brochosome systems in a number of recent studies (S. Yang, et al. Nat. Commun. 8:1285, 2017; L. Wang, et al., PNAS. 121: e2312700121, 2024). However, direct biological evidence or relevant field studies have been lacking to directly support the hypothesis that brochosomes are used for camouflage. This work provides the first biological evidence demonstrating that natural brochosomes can be used as a camouflage coating to reduce the leafhoppers' observability of their predators. The design of the experiments is novel.

We are extremely grateful for your positive feedback and insightful comments on our manuscript. We are delighted that you have recognized the efforts we have put into our

research on how brochosomes serve as a camouflage coating to reduce the detectability of leafhoppers to their predators. We have carefully considered your suggestions and have thoroughly revised the manuscript to address the shortcomings of the original version. We hope that the revised version meets with your approval. Below, please find our detailed point-by-point responses.

Weaknesses:

(1) The observation that brochosome coatings become sparse after 25 days in both male and female leafhoppers, resulting in increased predation by jumping spiders, is intriguing. However, since leafhoppers consistently secrete and groom brochosomes, it would be beneficial to explore why brochosomes become significantly less dense after 25 days.

Thank you very much for your valuable suggestions. We appreciate your interest in the reduction of brochosomal density on the surface of leafhoppers after 25 days. We believe that the primary reason for the decreased density of brochosomes on the leafhopper surface after 25 days is the reduced synthesis and secretion of brochosomes. The Malpighian tubules are the main sites for brochosome synthesis. As shown in Figure 2D and Supplementary Figure 1, the thick glandular segments of the Malpighian tubules in both male and female leafhoppers begin to atrophy 15 days after reaching adulthood. This indicates a gradual decline in brochosome synthesis and secretion after day 15 of adulthood. Following your suggestion, we have revised the discussion section of the manuscript to elaborate on this observation. The detailed changes can be found in lines 474-491 of the revised manuscript.

(2) The authors demonstrate that brochosome coatings reduce UV (specular) reflection compared to surfaces without brochosomes, which can be attributed to the rough geometry of brochosomes as discussed in the literature. However, it would be valuable to investigate whether the proteins forming the brochosomes are also UV absorbing.

Thank you very much for your valuable suggestions. Following your advice, we have successfully expressed four BSM genes in a prokaryotic system, purified the corresponding proteins, and applied them to quartz glass surfaces. We then measured the light reflectance of the quartz glass surfaces coated with these purified proteins. The results showed that the purified BSM proteins did not exhibit better antireflective properties compared to the control GST protein. For more details, please refer to Supplementary Figure 8 in the revised manuscript. We believe that the excellent antireflective properties of brochosomes are fundamentally due to their unique geometric shapes. The hollow pores within the brochosomes, with diameters of approximately 100 nm, are significantly smaller than most wavelengths in the visible spectrum. When light passes through these tiny pores, diffraction occurs, while light passing through the ridges of the brochosomes causes scattering. The interference between the diffracted and scattered light from these pores and ridges results in the observed extinction characteristics of brochosomes. We have incorporated these insights into the discussion section of the revised manuscript (lines 416-425 and lines 432-442 of the revised manuscript).

(3) The experiments with jumping spiders show that brochosomes help leafhoppers avoid predators to some extent. It would be beneficial for the authors to elaborate on the exact mechanism behind this camouflage effect. Specifically, why does reduced UV reflection aid in predator avoidance? If predators are sensitive to UV light, how does the reduced UV reflectance specifically contribute to evasion?

Thank you very much for your valuable suggestions. Based on your advice, we have included a detailed discussion on how reducing ultraviolet (UV) reflection can help insects avoid

predation. The revised content can be found in lines 445-460 of the revised manuscript.

“UV light serves as a crucial visual cue for various insect predators, enhancing foraging, navigation, mating behavior, and prey identification (Cronin & Bok, 2016; Morehouse et al., 2017; Silberglied, 1979). Predators such as birds, reptiles, and predatory arthropods often rely on UV vision to detect prey (Church et al., 1998; Li & Lim, 2005; Zou et al., 2011). However, UV reflectance from insect cuticles can disrupt camouflage, increasing the risk of detection and predation, as natural backgrounds like leaves, bark, and soil typically reflect minimal UV light (Endler, 1997; Li & Lim, 2005; Tovee, 1995). To mitigate this risk, insects often possess anti-reflective cuticular structures that reduce UV and broad-spectrum light reflectance. This strategy is widespread among insects, including cicadas, dragonflies, and butterflies, and has been shown to decrease predator detection rates (Hooper et al., 2006; Siddique et al., 2015; Zhang et al., 2006). For example, the compound eyes of moths feature hexagonal protuberances that reduce UV reflectance, aiding nocturnal concealment (Blagodatski et al., 2015; Stavenga et al., 2005). In butterflies, UV reflectance from eyespots on wings can attract predators, but reducing UV reflectance or eyespot size can lower predation risk and enhance camouflage (Chan et al., 2019; Lyytinen et al., 2004). Hence, the reflection of ultraviolet light from the insect cuticle surface increases the risk of predation by disrupting camouflage (Tovee, 1995)”

(4) An important reference regarding the moth-eye effect is missing. Please consider including the following paper: Clapham, P. B., and M. C. Hutley. "Reduction of lens reflection by the 'Moth Eye' principle." Nature 244: 281-282 (1973).

Thank you very much for pointing out the omission of the important reference on the “moth eye” effect. We sincerely apologize for the oversight. Based on your suggestion, we have now included the seminal paper by Clapham and Hutley (1973) in the revised manuscript. The reference has been added to both the Introduction and Discussion sections to provide a more comprehensive context for our discussion on anti-reflective structures in insects.

(5) The introduction should be revised to accurately reflect the related contributions in literature. Specifically, the novelty of this work lies in the demonstration of the camouflage effect of brochosomes using jumping spiders, which is verified for the first time in leafhoppers. However, the proposed use of brochosome powder for camouflage was first described by R.B. Swain (R.B. Swain, Notes on the oviposition and life history of the leafhopper Oncometopta undata Fabr. (Homoptera: Cicadellidae), Entomol. News. 47: 264-266 (1936)). Recently, the antireflective and potential camouflage functions of brochosomes were further studied by Yang et al. based on synthetic brochosomes and simulated vision techniques (S. Yang, et al. "Ultra-antireflective synthetic brochosomes." Nature Communications 8: 1285 (2017)). Later, Lei et al. demonstrated the antireflective properties of natural brochosomes in 2020 (C.-W. Lei, et al., "Leafhopper wing-inspired broadband omnidirectional antireflective embroidered ball-like structure arrays using a nonlithography-based methodology." Langmuir 36: 5296-5302 (2020)). Very recently, Wang et al. successfully fabricated synthetic brochosomes with precise geometry akin to those natural ones, and further elucidated the antireflective mechanisms based on the brochosome geometry and their role in reducing the observability of leafhoppers to their predators (L. Wang et al. "Geometric design of antireflective leafhopper brochosomes." Proceedings of the National Academy of Sciences 121: e2312700121 (2024)).

Thank you very much for your valuable suggestions regarding the revision of the introduction to accurately reflect the relevant contributions in the literature. Based on your feedback, we have thoroughly revised the introduction and added the suggested references to provide a comprehensive context for our study. The details of these revisions can be found in lines 84-94 of the revised manuscript.

Reviewer #3 (Recommendations for the authors):

(1) In Figure 2E, the data for Male-5d appears to be missing. Please verify and ensure all relevant data is included.

Thank you for pointing out the issue regarding the data presentation in Figure 2E. We apologize for any confusion caused by the overlapping data points and the less conspicuous color choice for Male-5d. We have carefully reviewed the data and confirmed that all relevant data points, including Male-5d, are indeed present in the dataset. In the revised manuscript, we have adjusted the color scheme for Male-5d and Female-5d in Figure 2E to ensure that both curves are clearly distinguishable, even in areas where they overlap. This adjustment should facilitate a more accurate and convenient observation of the data trends. We appreciate your attention to detail, and we believe these revisions have improved the clarity and readability of the figure.

(2) In Figure 6, please clarify the reflectance data in the inset. Clearly explain what the blue and light blue curves represent.

Thank you for your suggestion regarding Figure 6. We have revised the figure to improve clarity. The light blue curve now represents the reflectance measurements of leafhoppers with higher brochosome coverage, while the dark blue curve corresponds to those with lower coverage. These changes, along with updated labels in the figure legend, ensure that the data are clearly distinguishable and easy to interpret. We appreciate your feedback and believe these revisions have enhanced the overall clarity of the figure.

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